PROTOCOL OUTLINE

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1. Title

Development and Evaluation of an Explainable, Multimodal, and Temporal AI Framework for Early Student Dropout Prediction in Higher Education

2. Abstract

This section cannot be completed at this stage and will be developed in a later version, according to the teacher's instructions.

3. Introduction & Problem Statement

Student dropout remains a critical challenge in higher education, since institutions that identify at-risk students late have fewer real opportunities to intervene. In this study, dropout means that a student does not remain registered at the institution during the four regular academic periods following first registration. Universities now hold growing amounts of academic, administrative, and textual records over time, yet little work has examined how these data can be combined into an explainable and fair prediction approach for retention decisions (Alam, 2022; Glandorf et al., 2024; Phan et al., 2023; Quimiz-Moreira et al., 2025).

4. Literature Review

This section cannot be completed at this stage and will be developed in a later version, according to the teacher's instructions.

5. Questions, Hypotheses & Objectives

Research question: How can an explainable, multimodal, and temporal AI framework be designed and evaluated to improve early student dropout prediction and support transparent retention decisions in higher education?

General hypothesis: An explainable, multimodal, and temporal AI framework that integrates structured, textual, and longitudinal student data will achieve better predictive performance, temporal stability, and fairness than conventional static models based only on structured data.

Overall objective: To design and evaluate, within the doctoral period of three years, an explainable, multimodal, and temporal AI framework for early student dropout prediction in one higher education institution in Ecuador.

Specific objectives: (1) Integrate academic, administrative, textual, and temporal data into a reproducible dataset and modeling pipeline in Year 1; (2) develop and compare the framework with baseline models using predefined performance, calibration, fairness, and temporal validation metrics in Years 1 and 2; and (3) evaluate its interpretability and practical relevance and formulate transferable design principles in Year 3. The doctoral researcher will be responsible for all objectives, with institutional support for data access and domain validation.

6. Methodology

The study follows Design Science Research (Hevner et al., 2004), using CRISP-DM (Chapman et al., 2000) to structure the iterative design and evaluation of the framework within an embedded case study of one higher education institution in Ecuador. Deidentified historical academic, administrative, textual, and temporal records will be integrated, with dropout labeled from registration status during the four regular academic periods following first registration. The framework will be evaluated against baseline models using temporally separated training and test cohorts and ablation analysis, with metrics for predictive performance, calibration, fairness, temporal robustness, and explainability defined in advance and aligned with TRIPOD+AI reporting recommendations (Collins et al., 2024) and established XAI principles (Barredo Arrieta et al., 2020).

7. Ethical Considerations

Ethical approval and institutional authorization will be obtained before any data access, following the requirements of the UNMSM Research Ethics Committee, the CONCYTEC National Code of Scientific Integrity (CONCYTEC, 2024), and applicable Ecuadorian data protection regulations (Ecuador, 2021). Only the secondary data strictly necessary for the analysis will be pseudonymized, stored securely, and accessed by authorized personnel; the ethics committee will decide whether an exemption from individual consent is warranted. The study will also assess risks of bias, unfair predictions, reidentification, and incorrect interventions, and the resulting framework is meant to support, rather than replace, human judgment in retention decisions.

8. Expected Results

The study is expected to produce a validated framework whose dropout risk estimates are not only accurate, but also well calibrated, fair across student groups, and interpretable enough to support real decisions. Its contributions should include empirical evidence on combining multimodal and temporal data, a set of design principles that other institutions could adapt, and a working decision-support tool for timely retention actions.

9. Timeline & Budget

Year 1 will cover ethical approval, institutional authorization, data preparation, and baseline models; Year 2 will focus on framework development, testing, and temporal validation; and Year 3 will cover fairness and interpretability evaluation, refinement, publications, and thesis completion. The schedule includes a 25% contingency buffer, while the preliminary budget prioritizes secure storage, computing, software, dissemination, and publication, using institutional infrastructure when available.

10. Bibliography

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